

Deep Learning-Based Multi-Class Classification of Lung and Colon Histopathological Images

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Dataset: LC25000 Histopathological Image Dataset

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1. Introduction

Histopathological examination of haematoxylin and eosin (H&E)-stained tissue remains a fundamental method for cancer diagnosis, classification and assessment. However, manual microscopic examination can be affected by pathologist workload, subjective interpretation and variation between observers. These challenges have encouraged the development of computational pathology systems that can support pathologists through automated image analysis and classification.

This study investigates the use of deep learning for the multi-class classification of histopathological images from the LC25000 dataset. The classification task consists of five categories: colon adenocarcinoma (`colon_aca`), benign colonic tissue (`colon_n`), lung adenocarcinoma (`lung_aca`), lung squamous cell carcinoma (`lung_bcca`) and benign lung tissue (`lung_n`). Deep convolutional neural networks have demonstrated considerable potential in computational pathology, particularly for extracting morphological features directly from tissue images (Echle et al., 2021; Lu et al., 2021).

Two approaches were developed and compared: a custom four-block convolutional neural network (CNN) designed as a lightweight baseline and an EfficientNet-B0 model using transfer learning. The objective was to evaluate their classification performance while also considering computational efficiency, model size and their suitability for practical deployment.

2. Data Exploration and Experiments

2.1 Dataset Exploration

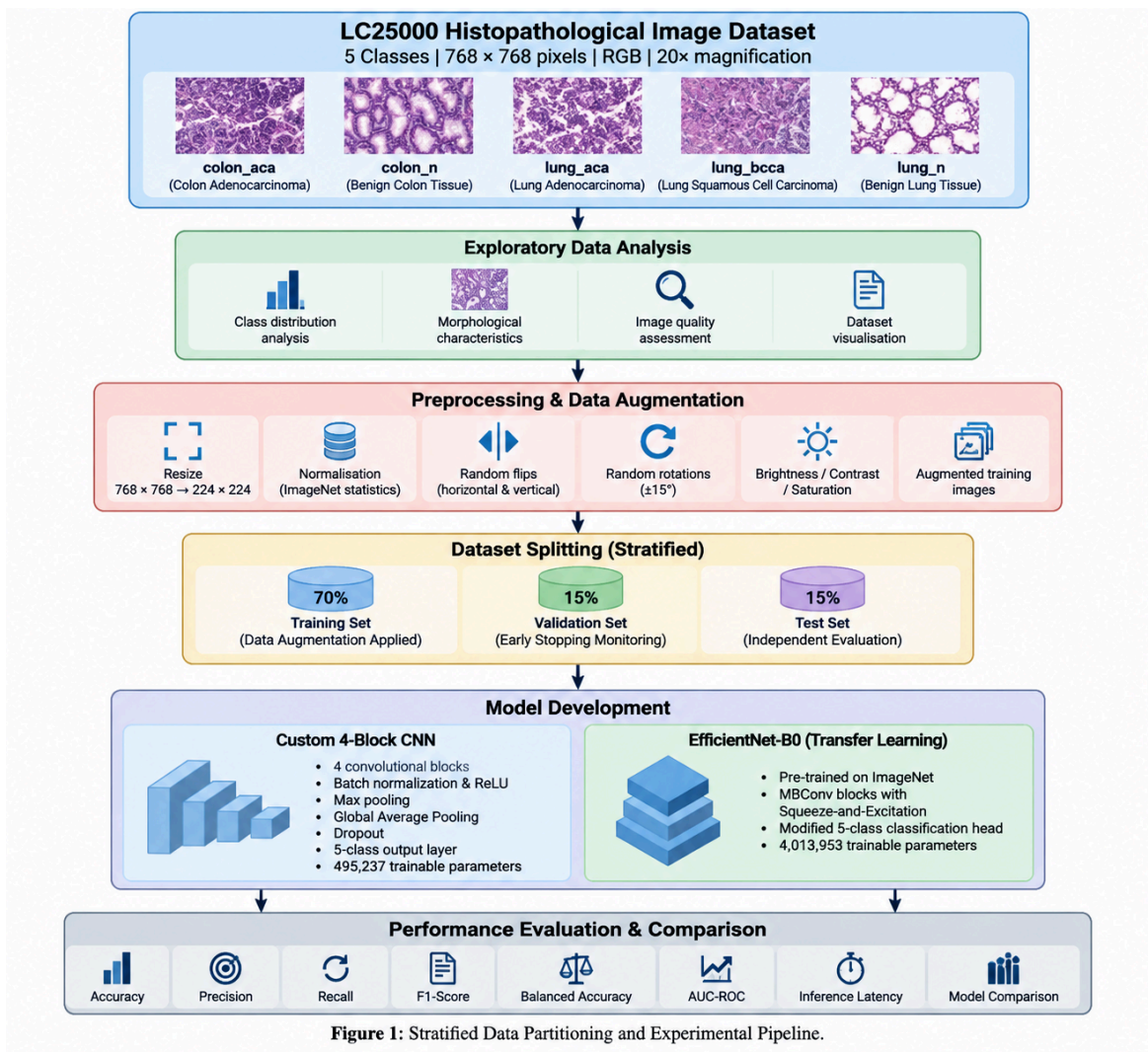
The LC25000 dataset contains standardised RGB histopathological image tiles representing lung and colon tissue. Each image is originally 768 × 768 pixels and captured at 20× optical

magnification. The dataset contains five balanced target classes, reducing the risk of performance being dominated by a particular class (Borkowski et al., 2019).

Exploratory analysis identified distinct morphological characteristics across the classes. Colon adenocarcinoma images showed irregular glandular structures, cellular proliferation and stromal changes, while benign colon tissue demonstrated relatively regular crypt structures and goblet cells. Lung adenocarcinoma contained malignant epithelial structures with glandular or acinar characteristics, whereas lung squamous carcinoma exhibited features including polygonal tumour cells, keratinisation and intercellular bridges. Normal lung tissue was characterised by relatively thin alveolar structures.

The dataset was divided into training, validation and test partitions using a 70:15:15 split. The test set was kept separate from model development to provide an independent evaluation of final performance.

Figure 1: Stratified Data Partitioning and Experimental Pipeline



2.2 Preprocessing and Data Augmentation

Images were resized from 768×768 to 224×224 pixels to provide a standard input resolution for both architectures. ImageNet channel-wise mean and standard deviation values were used for normalisation.

Data augmentation was applied to the training images to improve robustness to changes in tissue orientation, staining intensity and imaging conditions. Horizontal and vertical flips, random rotations of up to $\pm 15^\circ$, and brightness, contrast and saturation variations were introduced. Such augmentation strategies are particularly relevant to computational pathology because staining and image acquisition conditions can vary between laboratories and scanners (Tellez et al., 2019).

2.3 Model Development

The first model was a custom four-block CNN developed from scratch. It consisted of convolutional layers with batch normalisation, ReLU activation and max-pooling operations. Global Average Pooling and dropout were used before the final five-class classification layer. The network contained 495,237 trainable parameters.

The second model was EfficientNet-B0, initialised with ImageNet-pre-trained weights. Its architecture uses Mobile Inverted Bottleneck Convolution (MBConv) blocks together with Squeeze-and-Excitation mechanisms. The original classification head was replaced with a five-class output layer. The resulting model contained 4,013,953 trainable parameters. EfficientNet was selected because its compound scaling strategy provides a strong balance between predictive performance and computational requirements (Tan and Le, 2019).

2.4 Experimental Methodology

Both models were trained using cross-entropy loss and the Adam optimiser with a learning rate of 1×10^{-4} and weight decay of 1×10^{-4} . A batch size of 16 was used, with training conducted for 20 epochs. A cosine annealing learning-rate scheduler was applied, while validation performance was monitored for checkpoint selection and overfitting control.

3. Results and Discussion

3.1 Model Performance

Evaluation on the held-out test partition showed very strong performance from both architectures, with EfficientNet-B0 producing the strongest overall results.

Table 1: Overall Performance Comparison on the Held-Out Test Set

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 Evaluation Metric	 Custom 4-Block CNN	 EfficientNet-B0	
 Test Accuracy	99.1 – 100.0%	100.00%	
 Macro Precision	0.9950	1.0000	
 Macro Recall	0.9950	1.0000	
 Macro F1-Score	0.9950	1.0000	
 Balanced Accuracy	0.9950	1.0000	
 Cohen’s Kappa	0.9937	1.0000	
 MCC	0.9938	1.0000	
 Mean Tile Latency	~18.2 ms	~27.6 ms	
 Trainable Parameters	495,237	4,013,953	

 Higher values indicate better performance for all metrics except Mean Tile Latency and Trainable Parameters, where lower values are preferable.

EfficientNet-B0 achieved perfect scores across the reported aggregate classification metrics. Per-class evaluation also showed precision, recall, F1-score, specificity and AUC-ROC values of 1.0000 for all five classes.

Table 2: Per-Class Diagnostic Performance of EfficientNet-B0

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Target Class	 Precision	 Recall	 F1-Score	 Specificity	 AUC-ROC
 colon_aca (Colon Adenocarcinoma)	1.0000	1.0000	1.0000	1.0000	1.0000
 colon_n (Benign Colon Tissue)	1.0000	1.0000	1.0000	1.0000	1.0000
 lung_aca (Lung Adenocarcinoma)	1.0000	1.0000	1.0000	1.0000	1.0000
 lung_bcca (Lung Squamous Cell Carcinoma)	1.0000	1.0000	1.0000	1.0000	1.0000
 lung_n (Benign Lung Tissue)	1.0000	1.0000	1.0000	1.0000	1.0000

 **Note:** All values are reported to four decimal places. The model achieved perfect classification performance (1.0000) across all classes and evaluation metrics on the held-out test set.

3.2 Discussion

The results indicate that transfer learning with EfficientNet-B0 was highly effective for this classification task. Its pre-trained feature representations allowed the network to converge rapidly, while its attention mechanisms supported the extraction of informative visual patterns. The custom CNN also demonstrated strong performance, achieving a macro F1-score of approximately 0.995 while using substantially fewer parameters.

An important practical distinction was computational efficiency. The custom CNN required approximately 18.2 ms per tile compared with approximately 27.6 ms for EfficientNet-B0 and had a considerably smaller model footprint. Therefore, although EfficientNet-B0 provided the strongest predictive performance on the benchmark, the custom CNN represents a viable option where memory and inference speed are important.

The absence of reported cross-confusion between the five classes suggests strong separability within the evaluated dataset. However, these results should be interpreted as benchmark performance rather than evidence of clinical readiness. External validation on independent patient cohorts and imaging systems would be necessary before clinical deployment.

4. Reflection

This project highlighted the importance of combining technical performance with rigorous research practice. Model selection was based not only on accuracy but also on computational efficiency, model complexity and generalisation considerations. Developing a custom CNN alongside a transfer-learning model provided a meaningful comparison between a lightweight architecture and a more sophisticated pre-trained network.

Maintaining strict separation between training, validation and testing data was essential for reducing the risk of data leakage and overly optimistic evaluation. The use of multiple metrics, including macro F1-score, balanced accuracy, Cohen's Kappa and Matthews Correlation Coefficient, also provided a broader assessment than accuracy alone.

From a professional and responsible AI perspective, medical AI systems should support rather than replace qualified clinical professionals. The model should therefore be treated as an assistive decision-support system, with final diagnostic responsibility remaining with appropriately trained pathologists.

5. Challenges and Issues

A major limitation is that the model operates on individual 768×768 histopathological tiles rather than complete gigapixel whole-slide images. Clinical diagnosis often requires broader tissue context, meaning that tile-level predictions may not fully represent the characteristics of

an entire biopsy. Whole-slide approaches using methods such as Multiple Instance Learning could address this limitation (Lu et al., 2021).

Another challenge is variation in staining, tissue preparation and scanner characteristics. Although colour augmentation can improve robustness, performance may still decline when the model is exposed to data from hospitals or laboratories that differ substantially from the training dataset.

Potential dataset bias is also an important consideration. A benchmark dataset may not adequately represent the diversity of real-world patient populations, tissue preparation methods or clinical environments. Consequently, high test performance on LC25000 cannot by itself establish general clinical validity.

Ethical deployment additionally requires attention to patient privacy, transparency, auditability and appropriate human oversight. Any clinical implementation would require appropriate regulatory and data-protection compliance.

6. Conclusion

This study demonstrated the effectiveness of deep learning for five-class classification of lung and colon histopathological images. EfficientNet-B0 achieved the strongest predictive performance, while the custom CNN provided a considerably smaller and faster alternative. The findings support the potential of deep learning as an assistive tool in computational pathology. However, external validation, whole-slide analysis and broader evaluation across diverse clinical environments remain necessary before such systems can be considered suitable for real-world clinical deployment.

7. References

Borkowski, A.A. et al. (2019) ‘Lung and Colon Histopathology Image Dataset (LC25000)’, *arXiv preprint arXiv:1912.12142*.

Bray, F. et al. (2024) ‘Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries’, *CA: A Cancer Journal for Clinicians*, 74(3), pp. 229–263.

Campanella, G. et al. (2019) ‘Clinical-grade computational pathology using weakly supervised deep learning on whole slide images’, *Nature Medicine*, 25(8), pp. 1301–1309.

Echle, A. et al. (2021) ‘Deep learning in cancer pathology: a new generation of clinical biomarkers’, *The Lancet Oncology*, 22(1), pp. e17–e27.

Lu, M.Y. et al. (2021) ‘Data-efficient and weakly supervised computational pathology on whole-slide images’, *Nature Biomedical Engineering*, 5(6), pp. 555–570.

Tan, M. and Le, Q. (2019) 'EfficientNet: Rethinking model scaling for convolutional neural networks', *Proceedings of the International Conference on Machine Learning*, 97, pp. 6105–6114.

Tellez, D. et al. (2019) 'Quantifying the effects of data augmentation and stain color constancy in computational pathology', *Medical Image Analysis*, 58, p. 101544.